A straightforward method for wall impedance eduction in a flow duct

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The development of the advanced liner technology for aeroengine noise control necessitates the impedance measurement method under realistic flow conditions. Currently, the methods for this need are mainly based on the inverse impedance eduction principle, confronting with the problems of initial guess, high computation cost, and low convergence. In view of this, a new strategy is developed that straightforwardly educes the impedance from the sound pressure information measured on the duct wall opposing to the test acoustic liner embedded in a flow duct. Here, the key insight is that the modal nature of the duct acoustic field renders a summed-exponential representation of the measured sound pressure; thus, the characterizing axial wave number can be readily extracted by means of Prony's method, and further the unknown impedance is calculated from the eigenvalue and dispersion relations based on the classical mode-decomposition analysis. This straightforward method is simple in its basic principle but remarkably has the advantages of ultimately overcoming the drawbacks inherent to the inverse methods, incorporating the realistic multimode nonprogressive wave effects, high computational efficiency, possibly reducing the measurement points, and even avoiding the necessity of the duct exit impedance that bothers perhaps all the existing waveguide methods.
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I. INTRODUCTION

Acoustic treatments in the form of various liner structures have been demonstrated as the most effective means to control the noise of an aircraft engine over the past five decades and will continually play an important role in achieving the aggressive noise reduction goal of the next-generation quiet aeroengines. The new trend of higher bypass ratio turbofan design raises the formidable demand of greatly increasing noise attenuation with effectively diminished acoustically treated surface.[1] This is such a challenge that the current knowledge and techniques are inadequate; thus, major advancements are highly expected in future study.

To this end, the development of advanced liner technology fundamentally necessitates the progress in the impedance measurement method for nearly all the aspects, from improving the physical insights into the liner absorption mechanism to exploring the novel liner concepts such as the bias flow liner[2] and the active/passive hybrid types.[2] Moreover, more especially, driven by the need for increasing suppression efficiency, the acoustic liner optimization must be greatly enhanced over the current level. For this objective, a prediction model is essentially needed to determine the liner impedance as the crucial boundary condition for a propagation model to simulate the noise attenuation in the lined duct. The impedance model for such purpose not only depends on the experiment for validation but also must be highly empirical in its development process since a pure theory for practical impedance prediction would be a remote possibility.

However, subjected to the complicated high-speed flow condition in an aeroengine nacelle, reliable impedance measurement has long been proved a very difficult problem confronting the research community. Yet, it is well known that the grazing flow has considerable influence on the performance of an acoustic liner and, thus, its effect must be taken into account.[3] The enthusiasm and also the difficulty with this research topic can be evaluated from the numerous related papers and the various ideas being attempted over the past decades. A brief review of the representative inventions can be found in Ref. 4. Unfortunately, most of these have eventually been abandoned due to their limitations. Dean[4] made a successful improvement leading to the two-microphone *in situ* method that has gained wide application up to now. However, several problems have been reported with the use of this method. The *in situ* method requires the destructive installation of the microphones into the test liner, only provides the point-to-point measurement rather than the global impedance information that is more pertinent for the liner design purpose, and even fails when measuring the liners of multilayer structure or with the porous materials that have been extensively applied to meet the recent need for broadband noise control.[1]

To overcome the shortcomings of the *in situ* method, the inverse impedance eduction methods (also referred to as "inverse method" for brevity) have been developed and attracted wide research attention in recent years. This category of methods is based on the principle that the impedance of an acoustic liner strategically placed in a flow duct can be in-

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versely educed from minimizing the difference between the measured acoustic information at some characteristic points and the prediction of a forward propagation model. This concept, first put forward by Watson in the 1980s,[5,6] has recently evolved into many practical forms, in which the forward computation is done by the finite element[7–9] or mode-matching method,[10,11] and the measured acoustic information can be the sound pressure,[7–9] the reflection and transmission coefficients,[10,11] or the insertion loss.[12] However, all these methods are subjected to the general disadvantages. First, being relied on and sensitive to the initial guess, the convergence of the targeted searching is probably slow and even fails in the worse cases. Furthermore, since unique solution cannot be mathematically guaranteed except for very simple cases, the reliability of the inverse methods is compromised by the uncertainty whether the numerical searching converges to the true physical value or just a false result. Also, the computation cost is a major concern due to the repeated running of the numerical code in the iteration process, especially when the forward model is computationally intensive.

As for the inverse impedance eduction methods or perhaps all the existing waveguide methods in general, an inherent but not well-addressed weakness is the dependence of the methods on the duct exit reflective condition. Theoretically, the nonreflection condition is desired for these methods, but in practice this can only be nearly realized by employing a cumbersome anechoic duct termination, which is particularly inconvenient or even impractical for low-frequency measurement. Alternatively, the duct exit impedance is measured by the two-microphone method or similar techniques, but inevitably at the expense of bringing an additional source of error to the impedance eduction. It was found that in some situation, the inverse method is very sensitive to the duct exit condition and that a little variation in it causes a marked change of the educed impedance.[13] However, as far as we know, this negative influence on the impedance measurement accuracy has not been further analyzed in the previous investigations.

The limitations of the existing methods motivate the innovative research of impedance measurement methods under flow condition. It is the objective of this paper to develop a new method that straightforwardly educes the impedance of an acoustic liner strategically placed in a flow duct. In fact, the first attempt of straightforward impedance eduction in a flow duct is made by Armstrong *et al.*[14] even before the advent of the inverse methods. However, the so-called infinite waveguide method is based on the assumption of a single unidirectional propagating mode, too ideal to be satisfied in a physical experimental facility. Since then, confronted with the problem of how to consider the realistic acoustic environment containing multimode nonprogressive sound waves, little effort has been attempted along the direction of straightforward impedance eduction. Here, it is demonstrated that this tricky problem can be tackled by a methodology that takes full advantage of the modal nature of the duct acoustic field and draws inspiration from the application of the well-known Prony's method. Although extensively used to extract the wave numbers or propagation constants that are closely related to the absorption of various

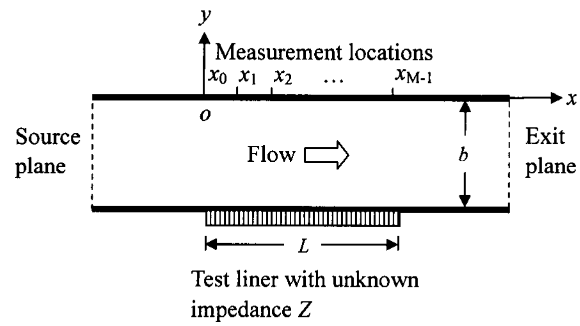

FIG. 1. Illustration of the straightforward impedance eduction in a flow duct.

waves,[15,16] the potential value of Prony's method in impedance measurement has received little attention so far in the field of acoustics. The present method is simple in its basic principle but has several striking advantages, for which the details are presented in the following context.

II. DESCRIPTION OF THE STRAIGHTFORWRD IMPEDANCE EDUCTION METHOD

As depicted in Fig. 1, a grazing flow environment is formed in a rectangular duct whose height is b. An acoustic liner of length L is placed in the lower central portion of the otherwise rigid duct wall. The test liner is treated as locally reacting with the unknown uniform impedance Z to be determined. It is reasonable to assume that the subsonic flow is uniform, with the Mach number being M_0. In the flow duct, when the time-harmonic plane sound wave generated in the hard wall section is incident upon the acoustic liner, multimode reflected and transmitted sound waves are excited. To collect the acoustic field information for the impedance eduction, the axial sound pressure distribution $p_u(x)$ is measured at a sequence of equally spaced locations on the upper-wall region directly over the acoustic liner. It can be seen that both the experimental facility and the implementation of the measurements are very similar to, but no more complicated than, the conventional inverse impedance eduction method.[7]

To facilitate the theoretical development, a coordinate system is introduced with x being along the duct axis and originated from the liner leading edge and putting $y=0$ on the upper duct wall. The harmonic sound propagation in the flow duct is governed by the convective Helmholtz equation for the sound pressure p,

$$\nabla^2 p - M_0^2 \frac{\partial^2 p}{\partial x^2} - 2ikM_0 \frac{\partial p}{\partial x} + k^2 p = 0, \tag{1}$$

where $i=\sqrt{-1}$, k is the free-space wave number, $k=2\pi f/c$, f is the frequency, and c is the sound speed. The boundary conditions at the rigid walls are that the normal particle velocity vanishes or, equivalently,

$$\frac{\partial p}{\partial y} = 0. \tag{2}$$

On the lined duct wall, the following impedance boundary condition is imposed,[17]

$$\frac{\partial p}{\partial y} = \frac{1}{Z}\left(ikp + 2M_0\frac{\partial p}{\partial x} + \frac{M_0^2}{ik}\frac{\partial^2 p}{\partial x^2}\right). \quad (3)$$

It can be intuitively deduced and mathematically proved that the uniformity of both the wall impedance and the incident sound pressure in the z direction renders an essentially two-dimensional sound field in the flow duct. A further analysis based on the classical mode-decomposition theory indicates that the sound pressure p inside the lined segment can be expressed in the following form:[10]

$$p(x,y) = \sum_{n=1}^{\infty} A_n^+ \cos(\beta_n^+ y)e^{-i\mu_n^+ x} + A_n^- \cos(\beta_n^- y)e^{-i\mu_n^- x}, \quad (4)$$

where $A_n^{\pm}$ are the complex modal amplitudes, and $\mu_n^{\pm}$ and $\beta_n^{\pm}$ are the complex axial and transverse wave numbers, respectively, which are related to each other through the dispersion relation

$$\mu_n^{\pm} = \frac{-M_0 k \pm \sqrt{k^2 - (1-M_0^2)(\beta_n^{\pm})^2]}}{1-M_0^2}. \quad (5)$$

Moreover, the transverse wave numbers are determined from the eigenvalue equation as

$$\frac{ik}{Z}\left\{\frac{1}{1-M_0^2}\left[1 \mp M_0\sqrt{1-\frac{1-M_0^2}{k^2}(\beta_n^{\pm})^2}\right]\right\}^2 = \beta_n^{\pm}\tan(\beta_n^{\pm} b). \quad (6)$$

Conversely, Eq. (6) provides a key relation to explicitly determine the unknown impedance from the transverse wave number and, thus, the axial wave number when combined with the dispersion relation.

It is obvious from Eq. (4) that on the upper duct wall where $y=0$, the sound pressure distribution $p_u(x)$ can be represented by a sum of complex exponentials,

$$p_u(x) = \sum_{n=1}^{N} A_n e^{-i\mu_n x}. \quad (7)$$

Note that the superscript "±" is omitted from the variables for simplicity, and for practical reason, the infinite summation has been truncated at the maximum mode number of N. A key insight of the present work is that, constrained by the measured data, the summed-exponential representation of $p_u(x)$ or Eq. (7) is deterministic, and the characterizing parameters μ_n and A_n can be identified by means of Prony's method. With the axial wave numbers being obtained, the impedance eduction is straightforwardly achieved by the simple algebraic manipulation of the dispersion relation (5) for transverse wave numbers and then the eigenvalue equation (6) for the unknown impedance.

Prony's method has extensively been used to extract the crucial parameters from both real-valued and complex-valued functions consisting of complex exponentials. For ease of reference, this method is outlined here. When the values of $p_u(x)$ are specified by the data measured at the M equal-spaced points, Eq. (7) turns to be

$$p_j = \sum_{n=1}^{N} A_n w_n^j \quad \text{for } j = 0,1,\ldots,M-1, \quad (8)$$

where $p_j = p_u(j\Delta x)$, $w_n = e^{-i\mu_n \Delta x}$, and Δx is the distance interval of the measurement points. Equation (8) can be further organized into $(M-N)$ consecutive sets of equations as

$$p_{s+r} = \sum_{n=1}^{N} A_n w_n^{s+r} \quad \text{for } s = 0,1,\ldots,N \text{ and } r = 0,1,\ldots,M-N-1. \quad (9)$$

It has been recognized that w_n is given by the roots of the Nth order polynomial equation

$$\sum_{s=0}^{N} C_s w^s = 0 \quad \text{with } C_N = 1. \quad (10)$$

Then, multiplying both sides of Eq. (9) by C_s and carrying out the summation over s, we get a set of linear algebraic equations for the coefficients of Eq. (10),

$$\sum_{s=0}^{N} p_{r+s} C_s = 0 \quad \text{for } r = 0,1,\ldots,M-N-1. \quad (11)$$

For solving Eq. (11), it is required that $M \geqslant 2N$ and the equal relation is used to keep M minimum for a fixed N in this paper. The pseudoinverse matrix method based on the singular value decomposition is employed to solve the involved linear equation sets, which is advantageous in handling the possible ill-condition issue of the matrix manipulation.

In principle, we can use any of the identified axial wave numbers to determine the unknown impedance. However, limited by the numerical accuracy of Prony's method, it is very important to choose the most appropriate wave number in practice. For this consideration, a simple and convenient selection criterion is proposed that the reliable wave numbers must be associated with the dominant modes with considerably larger amplitudes. Also, a systematic approach has been devised to differentiate the physical wave numbers from the spurious ones in Ref. 18. It is a big advantage of the present method that only one reliable axial wave number is sufficient to educe the unknown impedance, and it is readily found through the above identification and selection process. As expected, the picked-up wave number usually corresponds to the first-order forward traveling mode, in consistency with the way of introducing the sound wave into the flow duct.

It is worth noticing that, except for the algebraic dispersion relation (5) and eigenvalue equation (6), the present method is not burdened with the need for solving a complete duct propagation model, thus is computationally far more efficient than the inverse impedance eduction methods.

III. THE RESULTS AND DISCUSSIONS

In the present study, in order to validate the straightforward impedance eduction method (also referred to as "straightforward method" for brevity) via the widely accepted benchmark results, the experimental setup illustrated in Fig. 1 is taken to have the geometry identical to the NASA Langley Flow Impedance Test Facility[13] with the basic parameters being the duct height, 0.051 m, and the length of the test liner, 0.406 m. Some other conditions are that the sound speed is 344.3 m/s and the sound pressure level (SPL) at the source plane is kept to be 130 dB.

A. Straightforward impedance eduction from the FEM simulation of the flow duct experiment

For the initial validation purpose, a finite element model (FEM) duct propagation model is set up to simulate the acoustic experiment in the flow duct. To numerically solve the governing equation (1), two additional boundary conditions must be specified, i.e., at the duct inlet,

$$p = p_s, \tag{12}$$

where p_s is the sound pressure at the source plane, and at the duct exit,

$$\frac{\partial p}{\partial x} = -\frac{ikp}{M_0 + Z_e}, \tag{13}$$

where Z_e is the normalized specific acoustic impedance of the imaginary exit plane. In this paper, Z_e is either taken to be unity in a pure simulation or given by the NASA benchmark experiment when the test cases in Ref. 13 are considered. The present Galerkin FEM is similar to that of Ref. 7, but there is a major difference in the treatment of the impedance boundary condition at the acoustic liner. To incorporate into the FEM the second-order derivative term in Eq. (3), according to Ref. 7, cubic or higher-order basis function is necessarily used. Here, an alternative approach is adopted which reduces the derivative order of this term by the partial integration rule when the weak formulation of the FEM is written as follows:

$$\int_0^L \frac{M_0^2}{ikZ}\frac{\partial^2 p}{\partial x^2} N_m dx = \frac{M_0^2}{ikZ}\left(-\int_0^L \frac{\partial N_m}{\partial x}\frac{\partial p}{\partial x} dx + \left. N_m \frac{\partial p}{\partial x}\right|_0^L \right), \tag{14}$$

where N_m is the basis function. In Ref. 19, similar BC treatment method has already been proposed for the more general three-dimensional situation while discarding the term evaluated at the boundary of the lined area on the assumption of rapid but continuous impedance variation. However, the boundary evaluation term at $x=0$ and L is retained here, as shown by Eq. (14). So, based on Eq. (14), the linear basis function is employed to construct the FEM in this paper.

In Fig. 2, when $M_0=0.3$, $f=1.5$ kHz, $Z=2.0+i1.0$, and the duct exit is nonreflective, the FEM is compared to the mode-matching method that has been well established to represent the standard exact solution of the acoustic simulation.[10] The excellent agreement, as shown by the nearly overlapped solid and dash lines, readily lends the FEM being used to provide the simulation input for the present straightforward method. To best mimic the measurements, a random perturbation error is added to the FEM computation of the upper wall sound pressure as

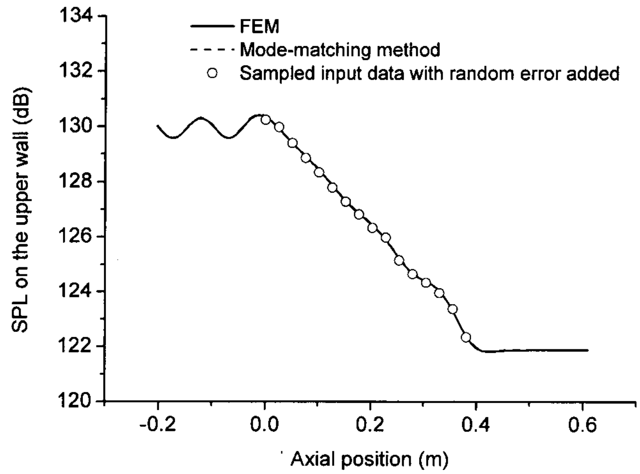

FIG. 2. FEM simulation of the upper-wall sound pressure compared to the standard mode-matching method. $M_0=0.3$, $f=1.5$ kHz, $Z=2.0+i1.0$, and $Z_e=1.0+i0.0$.

$$\hat{p}_u(x_k) = 10^{\Delta dB/20} p_u(x_k), \tag{15}$$

where $\Delta dB=(2N_R-1)\delta$, N_R is a random number between 0 and 1, and δ is set to the level of 0.1 dB typically experienced in the benchmark experiment.[8]

As the crucial step of the present method, Prony's method is employed to extract the axial wave numbers from the simulated upper-wall sound pressure that is uniformly sampled at a spacing of 0.0254 m over the x range from 0.0 to 0.381 m, as shown by the symbols in Fig. 2. The data sample range does not necessarily cover the whole length of the test liner. Actually, to avoid the contamination of the trivial evanescent modes due to the impedance discontinuities, the sound pressure data near the leading and trailing edges of the liner can be deliberately excluded from those used for the identification purpose.

For comparison purpose, the exact axial wave numbers are solved from the combination of Eqs. (5) and (6), and the corresponding modal amplitudes are computed from the mode-matching method of Ref. 10. Several approaches can be applied to solving the transcendental equation (6), such as the well-known Newton–Raphson, Eversman's integration,[20] and the newly homotopy methods,[21] but special care must be taken on the unfavorable possibilities of wrong initial guess, eigenvalue missing, etc. Because of this concern, the winding-number integral method[22] is recommended for the reliable calculation of transverse wave numbers in this paper.

In Table I, for the first two acoustic modes, the extracted axial wave numbers and the corresponding modal amplitudes show very good agreement with the exact results. So, as expected, the liner impedance eduction can be successfully accomplished by substituting either of the extracted μ_n into Eq. (5) for β_n and then using Eq. (6) to calculate Z. Relating to the dominant mode and thus being more reliable, $\mu_1=19.85-i2.299$ gives the result of $Z=1.993+i1.016$ that excellently reproduces the target value of $2.0+i1.0$.

B. Straightforward impedance eduction from the experimental data

With the success of the initial validation, it is very attractive to apply the present method to the situation of a real

TABLE I. Comparison between the exact and the extracted modal parameters from the FEM simulation shown in Fig. 2.

Mode number	Exact		Presently extracted	
	μ_n	A_n	μ_n	A_n
1^+	$19.87-i2.310$	$-31.06+i58.74$	$19.85-i2.299$	$-30.67+i58.35$
1^-	$-34.23+i5.276$	$-0.144-i0.013$	$-34.53+i4.886$	$-0.168+i0.018$

experiment. For years, the NASA Langley Flow Impedance Test Facility has provided benchmark data for the verification of various duct aeroacoustic simulation and measurements, particularly the inverse impedance eduction techniques. So, this experimental platform is also regarded as the touchstone of the present straightforward impedance eduction method. For this purpose, Ref. 13 is very helpful in publishing a comprehensive set of tabulated experimental data conveniently to be used by the present study. Although the sound pressure data given in the reference are distributed at almost evenly spaced measurement locations, this data format does not completely meet the requirement of using Prony's method. Hence, to prepare the input for the present method, a data preprocessing is performed with an interpolation scheme to create the equal-distance data sequence over the x coordinate range from 0.0003 to 0.3813 m. In this paper, if no specification is given, the data point number $M=16$ or, correspondingly, the spacing of 0.0254 m is chosen so that most of the sample points coincide with the original measurement locations, thus minimizing the interpolation error.

When there is no flow and the test frequency is 1.5 kHz, the measured sound pressure on the upper duct wall and the sampled input data are presented in Fig. 3. The effect of the higher-order modes can be assessed from the phenomenon that the sound pressure generally decays along the duct but with oscillations, which can hardly be treated by the infinite waveguide method requiring a perfect linear region on the SPL curve,[14] but brings little difficulty to the present method. From the input data, both the axial wave numbers and the

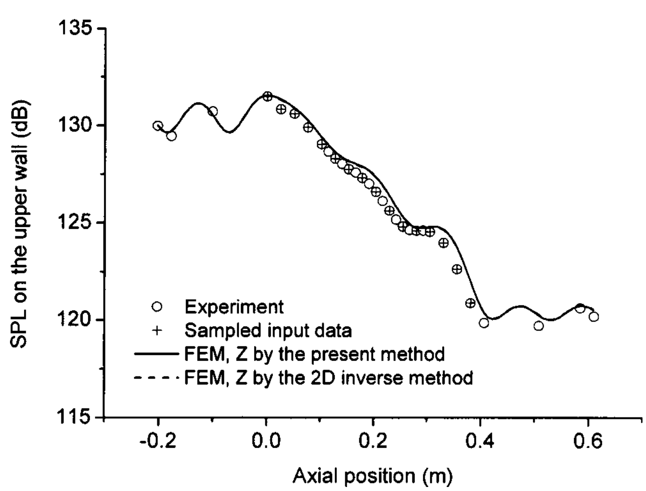

FIG. 3. The upper-wall sound pressure distribution presented by the experiment, the sampled input data, and the FEM simulations based on Z educed by the present method and the 2D inverse method of Ref. 13. $M_0=0.0$, $f=1.5$ kHz, and $Z_e=1.03+i0.08$.

modal amplitudes are extracted by Prony's method, as given in Table II. Note that the wave numbers are listed according to the magnitude of the modal amplitudes, so their sequences do not necessarily correspond to the orders of the real acoustic modes except for the most important situation that μ_1 is physically attributed to the 1^+ mode. It is also worth mentioning that in the experimental validation cases, the extracted wave numbers tend to be unreliable or even spurious as n increases due to their negligible influence and the numerical accuracy of Prony's method. So, evidently, the first extracted wave number is the correct choice for the impedance eduction purpose. In this case, with $\mu_1=22.88-i2.789$, we obtain $Z=0.98+i1.28$ from the simple calculation of Eqs. (5) and (6). This result is in very good agreement with that of the conventional two-dimensional (2D) inverse method,[13] $Z=1.02+i1.30$.

Furthermore, the present method is applied to the situation when flow is present. For $M=0.255$ and $f=1.5$ kHz, the measured SPL and the sampled input data are depicted in Fig. 4. Similarly, the axial wave numbers and the modal amplitudes are extracted and presented in Table II. From $\mu_1=19.06-i2.204$, the impedance is readily calculated to be $Z=1.12+i1.32$, showing good agreement with the previous result,[13] $Z=1.26+i1.26$. Another case is considered when the Mach number is up to 0.4 and the frequency is changed to 2.5 kHz. Although the experimental data exhibit larger oscillations at the liner edges due to the influence of higher speed flow, as shown in Fig. 5, the present method successfully extracts the key modal parameters (see Table II), and thus obtaining the liner impedance, $Z=0.89-i1.30$, that is quite near to $Z=0.75-i1.29$ from the 2D inverse method.[13] A thorough comparison between the present straightforward

TABLE II. The extracted axial wave numbers and the modal amplitudes for the first four significant modes at different Mach numbers.

M	f (kHz)	μ_n	A_n
0.000	1.5	$22.88-i2.789$	$55.14+i50.23$
		$6.723-i8.306$	$2.914-i0.600$
		$70.12-i29.57$	$0.918-i2.038$
		$-54.10-i32.15$	$-1.402-i1.576$
0.255	1.5	$19.06-i2.204$	$-18.31+i68.30$
		$4.794-i14.01$	$5.517+i3.735$
		$-87.74-i23.50$	$-0.755-i1.911$
		$-28.76+i4.140$	$0.975-i0.475$
0.400	2.5	$35.39-i2.173$	$54.71-i13.71$
		$-8.782-i28.49$	$27.10-i15.74$
		$-67.50-i20.20$	$-2.661+i3.766$
		$114.1-i2.059$	$0.053+i0.571$

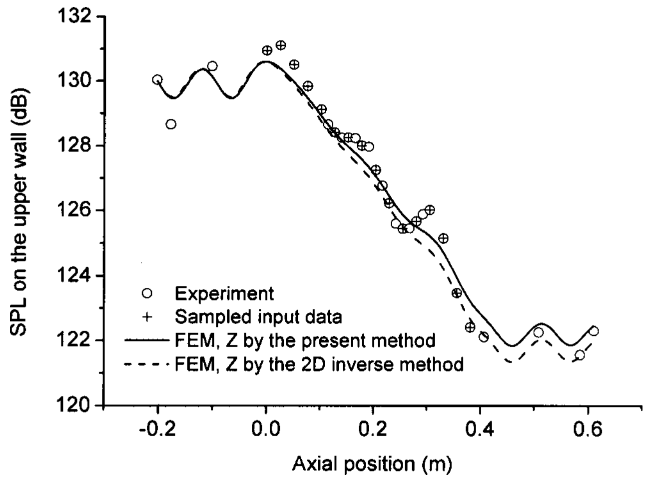

FIG. 4. The upper-wall sound pressure distribution presented by the experiment, the sampled input data, and the FEM simulations based on Z educed by the present method and the 2D inverse method of Ref. 13. M_0=0.255, f=1.5 kHz, and Z_e=1.10−i0.09.

TABLE III. Comparison between the present method and the previous 2D inverse method (Ref. 13) for the educed impedance at different Mach numbers when f=1.5 and 2.5 kHz, respectively.

	1.5 kHz		2.5 kHz	
M_0	Present	2D inverse	Present	2D inverse
0.000	0.98+i1.28	1.02+i1.30	1.43−i1.65	1.54−i1.60
0.079	1.00+i1.22	1.01+i1.25	1.37−i1.57	1.42−i1.39
0.172	1.07+i1.22	1.25+i1.18	1.17−i1.47	1.19−i1.55
0.255	1.12+i1.32	1.26+i1.26	1.04−i1.43	1.02−i1.46
0.335	1.05+i1.23	1.18+i1.27	0.98−i1.36	0.93−i1.43
0.400	1.07+i1.22	1.21+i1.08	0.89−i1.30	0.75−i1.29

method and the 2D inverse method of Ref. 13 is made for a variety of flow Mach numbers from 0 to 0.4 at two test frequencies, 1.5 and 2.5 kHz, respectively. As seen from Table III, the impedances educed by both methods agree well with each other. The occurrence of some discrepancies is understandable and can be explained in the way that the two educed impedances afford almost equally good FEM predictions of duct sound attenuation when compared to the measured data, as shown in Figs. 3–5 for the three typical flow speeds. Thus, the present straightforward method is satisfactorily validated for impedance measurement under practical flow condition.

The present method also reveals that although expected to be complicated in its spatial variations due to the presence of wall impedance discontinuities, the duct acoustic field can be adequately described with only a few predominant modes. Figure 6 shows that for the case of Fig. 4, the measured upper-wall sound pressure can be well reconstructed from summing the first four modal terms according to Eq. (7) with A_n and μ_n given in Table II. This simple truncated modal representation not only forms the basis for the present straightforward method but also provides valuable insights into the physical nature of the sound field under concern, and thus possibly leading to better understanding of the interaction between the acoustic liner and the duct sound waves in future study.

C. The influence of the number of the measurement points

As demonstrated in the previous subsections, the unknown impedance can be straightforwardly educed using 16 input data points, nearly as half as that used by the inverse method of Ref. 13. In this subsection, we will further explore the possibility of educing the impedance with a relatively small number of measurement points. This is very important since the reduction of measurement points not only means increasing the measurement efficiency but also reducing the sources of error and thus improving the reliability. For the acoustic phenomenon under consideration, the simplicity of the incident sound wave renders that there are only one or a few dominant propagating modes in the flow duct. In this situation, Prony's method can be very effective and the crucial wave numbers can be precisely extracted but not necessarily using densely distributed data points.

Here, the case shown in Fig. 4 is re-examined. When reducing the measurement points, the input data of the sound

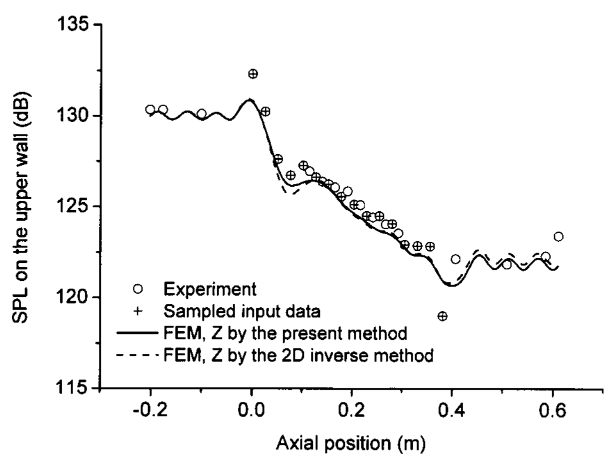

FIG. 5. The upper-wall sound pressure distribution presented by the experiment, the sampled input data, and the FEM simulations based on Z educed by the present method and the 2D inverse method of Ref. 13. M_0=0.4, f =2.5 kHz, and Z_e=0.92−i0.14.

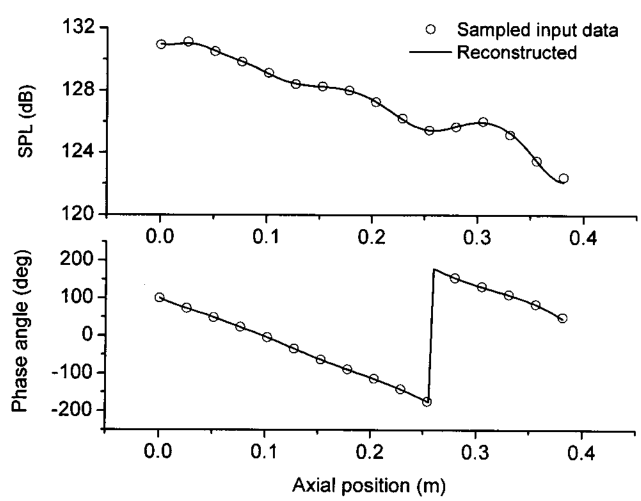

FIG. 6. The upper-wall sound pressure reconstructed from Eq. (7) with μ_n and A_n given in Table II. M_0=0.255, f=1.5 kHz, and Z_e=1.10−i0.09.

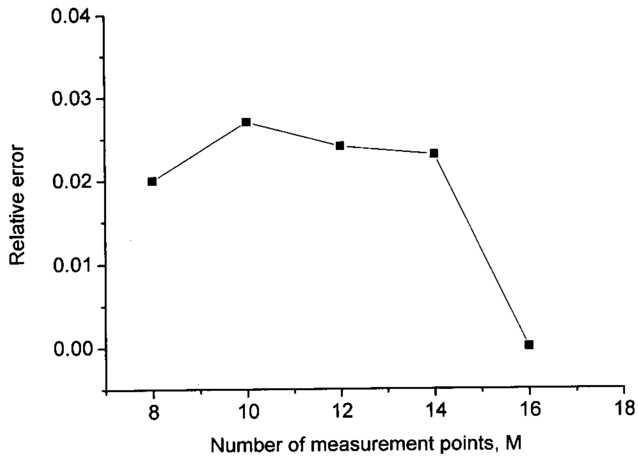

FIG. 7. The influence of the number of the measurement points on the educed impedance. $M_0=0.255$ and $f=1.5$ kHz.

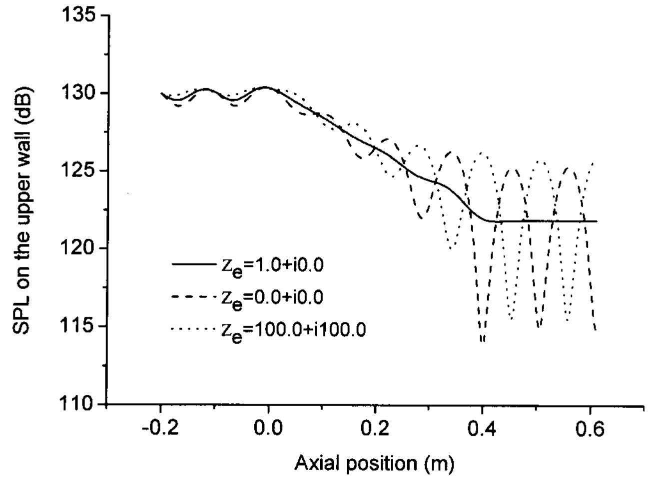

FIG. 8. The variation of the upper-wall sound pressure distribution subjected to different duct exit impedances. $M_0=0.3$ and $f=1.5$ kHz.

pressure are obtained by means of the interpolation method described in Sec. II B. Figure 7 shows that we can use as few as eight measurement points to satisfactorily educe the liner impedance with a relative error of less than 3% defined by $|Z-Z_0|/|Z_0|$, where Z_0 is the result for $M=16$. On the other hand, the reduction of the measurement points is limited by the aliasing effect in Prony's method. In order to avoid the multivalue problem when determining μ_1 from the relation $w_1=e^{-i\mu_1\Delta x}$, the distance interval must meet the requirement

$$\Delta x < \frac{2\pi}{\mathrm{Re}(\mu_1)}. \tag{16}$$

The above relation is derived using the condition that $\mathrm{Re}(\mu_1)$ is always positive. So, if $\mathrm{Re}(\mu_1)$ is approximately taken to be $k/(1+M_0)$ with $M_0=-0.4$ and $f=3400$ Hz given by the cutoff frequency of the rectangular duct, we can estimate that the smallest distance interval is 0.06 mm or, correspondingly, the minimum number of the measurement points is 8. A common principle for choosing M and N would be very difficult and beyond the scope of the present work. However, a numerical survey over the experimental cases in Ref. 13 indicates that the modal expansion can be effectively truncated at $N=2$ or 3 and generally using eight measurement points is sufficient to satisfactorily educe the unknown impedance.

It is further discussed that, as for the issue of reducing the measurement points, the difference between the conventional inverse impedance eduction methods and the present method lies in whether a direct or a parametric fitting is adopted. The direct fitting of the inverse methods is based on a sort of black box approach and thus necessitates dense data points to follow the variation of sound pressure curve as exactly as possible in order to minimize the residual error. By comparison, best using the modal representation of duct sound propagation as prior knowledge, the present straightforward method fits a model equation to the measured data and therefore is able to extract the crucial parameters by means of fewer measurement points.

D. The influence of the duct exit reflective condition

It is a striking feature that the present method is independent of the duct exit reflective condition, as inferred from its basic principle. In contrast, whether the previous infinite waveguide method or inverse methods, they are based on a full duct propagation model, thus requiring the complete avoidance or the accurate determination of the duct exit reflection. However, this is by no means an easily treated problem when there is flow present, as discussed in the Introduction.

The benchmark experiment on the NASA Langley Flow Impedance Test Facility[13] is conducted only on the nearly anechoic duct exit condition. Because of lacking experimental data, the FEM simulation is used to provide the input data for the validation purpose in this subsection. For the case of $M_0=0.3$ and $f=1.5$ kHz, as plotted in Fig. 8, when the duct exit impedance varies within a wide range, the upper-wall sound pressure level exhibits a marked change in its distribution along the duct, showing strong standing wave pattern under the highly reflective exit conditions. Regardless of the changes in Z_e, the present method can educe the unknown impedance in the same accuracy, as indicated by the results in Table IV. Actually, the present straightforward method essentially has no need for the information of the duct exit reflection.

IV. SUMMARY

In this paper, to overcome the limitations of existing methods for impedance measurement under practical flow

TABLE IV. The extracted axial wave numbers and educed impedance when the duct exit impedance has different values.

	Exit impedance Z_e		
Results	$1.0+i0.0$	$0.0+i0.0$	$100.0+i100.0$
μ_n	$19.85-i2.299$ $-34.53+i4.886$	$19.86-i2.300$ $-34.06+i5.166$	$19.87-i2.337$ $-35.19+i5.442$
Z	$1.993+i1.016$	$1.997+i1.013$	$1.989+i0.981$

condition, a new straightforward strategy is developed to educe the impedance of an acoustic liner embedded in a flow duct. Here, the basic principle is that the modal nature of the duct acoustic field renders a summed-exponential representation of the sound pressure measured on the duct wall; thus, the characterizing axial wave number can be readily extracted by means of Prony's method, and further the unknown impedance is straightforwardly calculated from the combination of the eigenvalue equation and the dispersion relation. At current stage, an assumption of uniform flow remains made but it is reasonable for most of the practical applications.

The present method is simple but remarkably has the following advantages.

(1) Being a straightforward method, it ultimately avoids the various problems of initial guess dependence and low or even wrong convergence that are inherent to the inverse methods.
(2) Except for the algebraic manipulation of the dispersion relation and the eigenvalue equation, it needs neither the complete solution of a duct propagation model nor any iteration computations; thus is numerically very efficient.
(3) Based on the parametric rather than the direct fitting approach, it shows the possibility of educing the liner impedance with considerably reduced number of measurement points compared to the inverse impedance eduction methods.
(4) Different from the early attempt of the straightforward impedance eduction or the so-called infinite waveguide method, it considers the realistic acoustic environment containing the multi-mode nonprogressive sound waves in the flow duct, without making any assumption on the modal contents of the acoustic field.
(5) It unexpectedly circumvents the necessity of the duct exit boundary condition that causes trouble to the existing waveguide methods for either using a cumbersome anechoic termination or accuracy degeneration due to the errors in the exit impedance measurement.

It is well anticipated that, having been validated by the well-accepted benchmark experiment, this novel method will demonstrate its potential benefits in the future study and applications.

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